

Isopropyl Alcohol Production

Submitted to: Prof. Jonathan Verrett & Prof. Emily Cranston

Submitted by: Avery Lai, Christine Agustin, Shivom Rathore, Yanya Yang

Courses: CHBE 220 & CHBE 201

Date: December 8, 2024

EXECUTIVE SUMMARY

A client in Tokuyama, Japan wants to produce 100,000 tonnes of isopropyl alcohol (IPA) annually at their plant. IPA is used in antifreeze compositions, industrial solvents, cosmetic products, in the manufacture of acetone, glycerol, and isopropyl acetate, and as a chemical intermediate in solvent applications within the medical field. Worldwide production was 2.3 million tonnes in 2022. The gross economic potential for this process is estimated to be \$59.4 million US dollars annually.

Direct hydration at a high temperature and pressure is recommended because it has 98-99% selectivity and 99.99% purity of IPA, maximizing IPA yield for large-scale production with minimal by-products. Separation is achieved by extractive distillation. Unconverted propylene is recycled to the reactor.

Energy transfer in IPA production occurs following the reactor, where the product stream is cooled from 250°C to its dew point (81.7°C) before it enters the first distillation column. The cooling utility selected is cooling water at 25°C. The total cooling utility needed for this stream is 7.173 GJ/h. The cost of cooling water based on this number is \$2.71/h or \$21,690.30/year, assuming 8,000 hours of operation annually. The amount of cooling water needed for this process is 338 tonnes/h.

Propylene and water are heated to 250°C and compressed to 150 atm before entering the reactor. The fluid exiting the reactor is cooled and depressurized before entering the first distillation column. This produces a vapor mixture of propylene and diisopropyl ether (DIPE), and a liquid consisting of the IPA-water azeotrope. The vapor stream enters a condenser and reflux drum. The liquid stream is recycled to the distillation column.

The liquid IPA stream is split into two. One stream is sent to a reboiler, and the other is processed in an extractive distillation column.

The control strategy focuses on controlling the temperature and pressure of propylene before it enters the reactor. Temperature control ensures propylene enters the reactor at 250°C. Pressure control ensures propylene enters the reactor at 150 atm.

RECOMMENDATION

Direct hydration at high temperature and pressure is recommended for producing 100,000 tonnes per year of IPA for a plant located in Tokuyama, Japan. With high selectivity and minimal environmental impact, making it the recommended process.

Table of Contents

EXECUTIVE SUMMARY	ii.
LIST OF TABLES	vi.
LIST OF FIGURES	vi.
1.0 INTRODUCTION FOR DESIGN REPORT ON ISOPROPYL ALCOHOL PRODUCTION	1
1.1 Introduction.....	1
1.2 Purpose.....	1
1.3 Significance.....	1
2.0 REACTION PATHWAYS	3
2.1 Introduction.....	3
2.2 Reaction pathway 1: hydrogenation of acetone	3
2.3 Reaction pathway 2: indirect hydration of propylene	3
2.4 Reaction pathway 3: direct hydration of propylene	4
2.5 Recommended pathway	4
3.0 SEPARATION TECHNIQUE	5
3.1 Introduction.....	5
3.2 Reaction pathway	5
3.3 Separation strategy	5
4.0 ENERGY BALANCES.....	8
4.1 Introduction.....	8
4.2 Energy transfer: cooling the stream before separation.....	8

4.3 Utility stream, heat capacities and operating costs	8
5.0 PROCESS DESCRIPTION AND FLOW DIAGRAM (PFD)	10
5.1 Introduction.....	10
5.2 Reacting water and propylene to produce isopropyl alcohol	10
5.3 Separating isopropyl alcohol-water azeotrope and byproducts.....	10
6.0 CONTROL STRATEGY	11
6.1 Introduction.....	11
6.2 Unit controlling temperature.....	11
6.3 Unit controlling pressure.....	11
7.0 FINDINGS AND RECOMMENDATIONS.....	12
REFERENCES	13
APPENDICES	15
Appendix A: Economics calculations	15
Appendix B: Material balance sample calculations	16
Appendix C: Dew point and energy balance calculations.....	17
Appendix D: Process flow diagram	20
Appendix E: Control strategy process flow diagram	22

List of Tables

Table 1. Common properties for compounds in isopropyl alcohol production.....	2
Table 2. Flows into and out of the reactor based on production of 12.5 tonnes/h of isopropyl alcohol ...	6
Table 3. Properties relevant for separation of each compound	6
Table 4. Key parameters from dew point calculations using Raoult's law	9
Table D1. Stream table for the reactor and distillation column	21

List of Figures

Figure 1. IPA structure	1
Figure 2. IPA process input-output diagram	2
Figure D1. Process flow diagram for reactor and distillation column in isopropyl alcohol production	21
Figure E1. Process flow diagram showing controls for the propylene heat exchanger and compressor	22

1.0 INTRODUCTION FOR DESIGN REPORT ON ISOPROPYL ALCOHOL PRODUCTION

1.1 Introduction

A design report on the production of isopropyl alcohol (IPA) for a client near Tokuyama, Japan is presented. Given its versatility, production of IPA is increasing in the Asia-Pacific region to accommodate its growth. This memo outlines the geographical factors influencing IPA production, the synthesis pathway, and the gross economic potential of the product. An input/output diagram and a table of key compounds and their prices are included.

1.2 Purpose

Our client is requesting a process to produce 100,000 tons per year of IPA at a minimum purity of 99.99%. The proposed location for the plant is near Tokuyama, Japan, which has highs in the summer of 26°C and lows in the winter of 2°C [1].

1.3 Significance

IPA (Figure 1) is a volatile liquid that is used primarily as a chemical intermediate in solvent applications within the medical field. IPA has a diverse range of applications, including use in antifreeze compositions, industrial solvents, cosmetic products, and in the manufacture of acetone, glycerol, and isopropyl acetate [2]. IPA is generally produced commercially in large plants; 2.3 million tonnes were produced in 2022 [3].

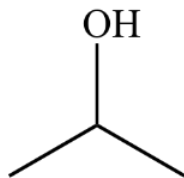
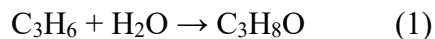


Figure 1: IPA structure, figure reproduced from [4].

Direct hydration of propylene under high temperature and pressure is the most efficient and economical process with the least environmental impact [5]. Liquid propylene C_3H_6 is heated and fed into a reactor under pressure, where it reacts with water to form IPA, with the aid of aqueous tungsten oxide catalysts. The reaction is as follows:



The gross economic potential for this process is estimated to be \$59.4 million US dollars annually based on the prices in Table 1, with input and output flows (Figure 2).

Calculations and citations for sources for chemical pricing are included in appendix A.

Table 1: Common properties for compounds in isopropyl alcohol production.

Name	Formula	Molecular weight (g/mol)	Price (\$ USD/tonne)
Propylene	C_3H_6	42	816 [6]
Water	H_2O	18	1
IPA	C_3H_8O	60	1165 [7]

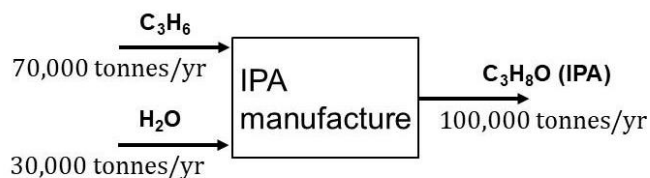


Figure 2: IPA process input-output diagram.

2.0 REACTION PATHWAYS

2.1 Introduction

Three reaction pathways for manufacturing IPA are described in terms of their yield, cost, and environmental factors. The direct hydration process is recommended. Indirect hydration, direct hydration, and hydrogenation of acetone processes are described below.

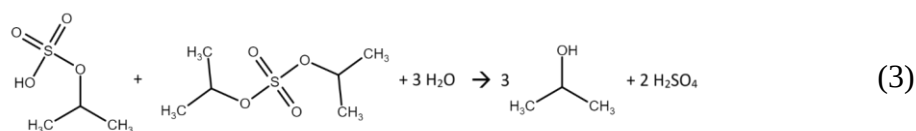
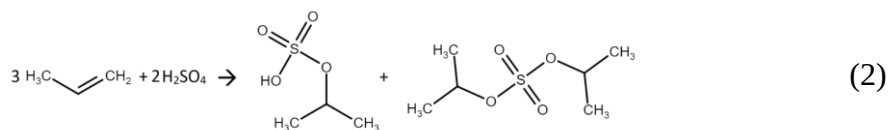
2.2 Reaction pathway 1: hydrogenation of acetone

The hydrogenation of acetone is a two-step pathway whereby acetone is reacted with hydrogen gas in the presence of a catalyst to produce IPA (reaction 1) [8]. A minor by-product and a heavy ends by-product are formed [9]. IPA is then distilled to obtain a high purity product with a 99.99% yield. Depending on the catalyst, required temperature ranges from 150-170°C and selectivity ranges from 98-100% [10]. The process is environmentally friendly due to the use of acetone, a widely available by-product of other processes [11]. This process is best suited for small-scale production because it is the most expensive [8].



2.3 Reaction pathway 2: indirect hydration of propylene

Weak acid indirect hydration of propylene is also a two-step pathway. Sulfuric acid is added to propylene to produce a mixture of isopropyl hydrogen sulfate and diisopropyl sulfate (reaction 2). Sulfates are then hydrolyzed to IPA and sulfuric acid in reaction (3). The sulfuric acid is recycled and used in reaction (2) at 60% concentration [12]. Byproducts include DIPE, hydrocarbons and sulfur oxide pollutants.



2.4 Reaction pathway 3: direct hydration of propylene

Direct hydration of propylene uses a highly active catalyst system with water at high temperature (240°C -270°C) and high pressure (150-250 atm) [13]. Propylene reacts with water to form IPA, reaction (4), with only minor by-products such as DIPE and n-propanol. This process eliminates the need for handling large amounts of sulfuric acid and is more cost-effective for large-scale production. Reaction conditions result in higher selectivity of IPA.



Reaction (4) favours the formation of IPA at lower temperatures (130°C-160°C) due to equilibrium conditions. However, higher reaction rates are achieved by operating at higher temperatures and the use of a selective catalyst. Aqueous phosphotungstic acid is the chosen catalyst because it is highly active and resistant to hydrolysis. Reaction (4) is exothermic with a heat of reaction of 12 kcal/mol [13].

The direct hydration process uses a high-pressure reactor to ensure good contact between the propylene and the water/catalyst phase. The reactor operation is continuous, with propylene gas bubbled through the aqueous catalyst solution. The process achieves high per-pass conversions of propylene of 60-70%, 99.99% purity of IPA, and IPA selectivities of 98-99% [13]. After the reaction, separation occurs on the product stream. Unconverted propylene is separated and recycled back to the reactor. Then, a series of distillation steps is used to separate IPA from water and minor by-products. To break the IPA-water azeotrope and produce high-purity IPA, azeotropic distillation is used.

2.5 Recommended pathway

Direct hydration of propylene under high-temperature and high-pressure conditions is recommended because of its efficiency. With 98-99% selectivity and 99.99% purity of IPA, this process maximizes IPA yield for large-scale production. Minimal by-products in this process makes it environmentally preferable.

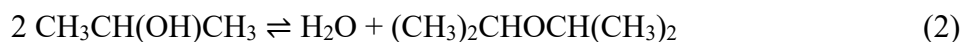
3.0 SEPARATION TECHNIQUE

3.1 Introduction

Details are provided regarding a separation unit to be used in a process to produce IPA. A separation strategy for the product stream from the hydration reaction is described. A material balance for the flow of materials entering the reactor unit is provided based on the reaction and anticipated product flow rate. Sample calculations are enclosed.

3.2 Direct hydration process for manufacturing isopropyl alcohol

Direct hydration of propylene at high temperature (240°C -270°C) and pressure (150-250 atm) is selected to produce IPA [13]. In the presence of a phosphotungstic acid catalyst, propylene reacts with water to form IPA following reaction (1). IPA further undergoes etherification to form DIPE as a minor byproduct in reaction (2).



Reaction (1) achieves a high per-pass conversion of propylene of 60-70% and is 98-99% selective for IPA [13]. This process uses a continuous high-pressure reactor, with propylene gas bubbled through the aqueous catalyst solution.

3.3 Separation strategy for isopropyl alcohol and water using extractive distillation

A conversion rate of 60% is assumed [13]. The product stream contains unreacted propylene, azeotropes of IPA and water, and it is assumed that DIPE is the only byproduct and is negligible.

Chemical plants commonly assume 8,000 hours of operation annually. Applying this operational capacity, the flow of IPA is 12.5 tonnes/h based on 100,000 tonnes of production annually. A mole and mass balance based on this flowrate with a 60% conversion in the reactor results in the values for the exit stream of the reactor unit shown in Table 1 with sample calculations provided in Appendix B.

Table 2: Flows into and out of the reactor based on production of 12.5 tonnes/h of IPA.

Compound	Formula	Molecular weight (g/mol)	Flow into reactor (tonnes/h)	Flow out of reactor (tonnes/h)
Propylene	CH_3CHCH_2	42.081	14.6	5.85
Water	H_2O	18.015	6.25	2.49
IPA	$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$	60.100	-	12.5
DIPE	$(\text{CH}_3)_2\text{CHOCH}(\text{CH}_3)_2$	102.172	-	negligible

IPA forms an azeotrope with water. The IPA-water azeotrope is separated from unreacted propylene and DIPE, then IPA is extracted from water using ethylene glycol (EG).

Relevant physical properties for separation of each compound are shown in Table 2.

Table 3: Properties relevant for separation of each compound.

Compound	Boiling point (°C)	Heat of vaporization (kJ/mol)	Melting point (°C)	Heat of fusion (kJ/mol)
Propylene [14]	47.7	18.31	-185.2	3.0
Water [15]	100.0	40.7	0.0	6.0
IPA [16]	82.6	39.85	-84.5	5.4
Water-IPA azeotrope [17]	80.3	-	-	-
DIPE [18]	68.4	50.44	-85.7	12.0
EG [19]	197.6	49.6	-13	11.2

The mixture initially exits the reactor at a temperature of 250°C and the stream is rapidly cooled to begin the distillation process [20], as shown in Figure (1). Separation guidelines [20] make distillation a good choice for separating these fluids to achieve purity in all product streams. Separation guidelines advise separating the largest product stream first

and using distillation to remove the product with the highest heat of vaporization first if possible. All separation columns are assumed to be operating at atmospheric pressure.

The IPA-water azeotrope is first separated from DIPE and propylene because it has the highest boiling point. It leaves the first distillation column in the heavy stream, which operates between 65°C and 80°C. The light stream consisting of DIPE and propylene enters a second distillation column operating at temperatures between -47°C and 65°C. DIPE in the heavy stream is used as fuel for the reactor, while propylene in the light stream is recycled back to the reactor. The IPA-water azeotrope enters an extractive distillation column operating between 80°C and 100°C, where EG is added as an entrainer and breaks the azeotrope. EG is not volatile and interacts strongly with water through hydrogen bonds, effectively separating water from IPA [17]. Purified IPA is collected in the product stream. EG and water enter a vacuum distillation column where water exits in the light stream and EG in the heavy stream is recycled as an entrainer [17].

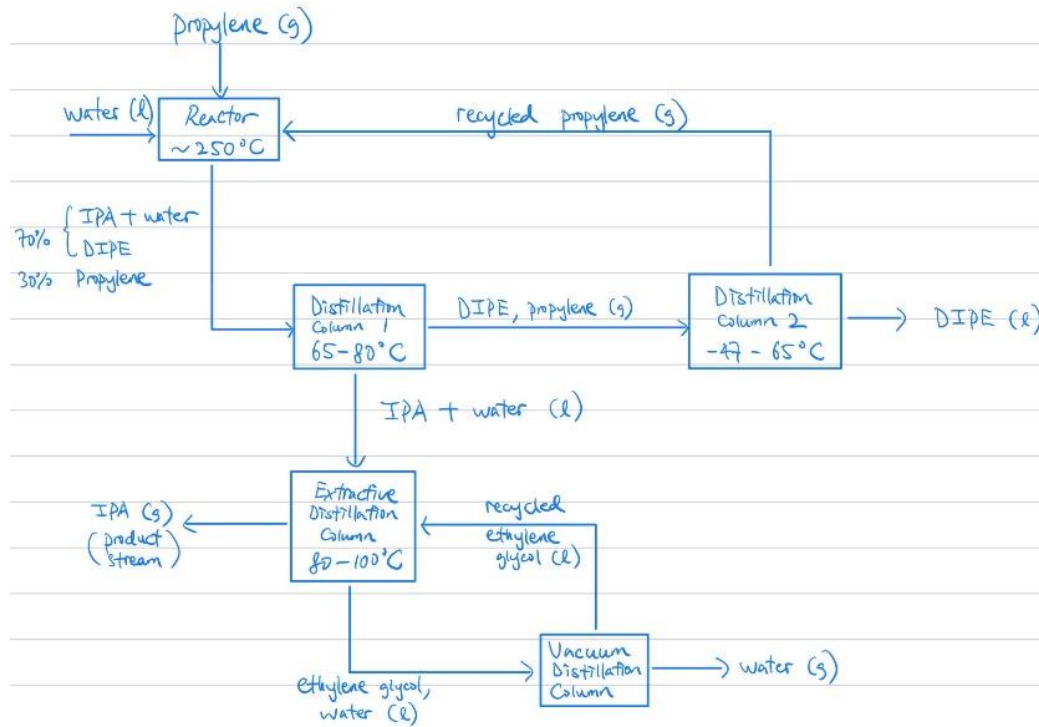


Figure 3: Flow diagram of the separation process.

4.0 ENERGY BALANCES

4.1 Introduction

Energy transfer in IPA production downstream of the reactor is described, specifically for where the stream is fed into the first distillation column to separate the IPA-water azeotrope from DIPE and propylene. The best utility to cool this stream is cooling water. The amount of energy that is exchanged for cooling is calculated as well as the total cost of the utility per hour and for a year of operation. Sample calculations are enclosed.

4.2 Energy transfer: cooling the stream before separation

Energy transfer in IPA production occurs following the reactor, where the stream is fed into the first distillation column to separate the IPA-water azeotrope from DIPE and propylene. The stream needs to be close to the vapour-liquid equilibrium (dew point) to feed it into the distillation column. The dew point is 81.7°C, which is calculated using Raoult's law. The results of key calculations are shown in Table 1.

4.3 Utility stream, heating capacities and operating costs

Cooling water is the best utility to cool this process stream. The desired exit process stream leaves the reactor at 250°C. Cooling water is at 25°C, which is more than a 10°C difference from the process stream, so it has sufficient driving force for energy transfer.

Heat capacities for the gas components are estimated to find out how much of this utility is required to cool the process stream. The heat capacities for propylene, IPA, and water vapour were calculated at a temperature range of 355 K – 523 K.

The total cooling utility needed for this stream is 7.173 GJ/h. The cost of cooling water based on this number is \$2.71/h or \$21690/yr, assuming 8,000 hours of operation. The amount of cooling water needed for this process is 338 tonnes/h.

Table 4: Key parameters from dew point calculations using Raoult's law.

Compound	Formula	Flow to separate (tonne/h)	Flow to separate (tonne-mol/h)	y_i (gas mole fraction)
Propylene	CH ₃ CHCH ₂	5.85	0.139	0.287
Water	H ₂ O	2.49	0.138	0.285
IPA	CH ₃ CH(OH)CH ₃	12.5	0.208	0.429

5.0 PROCESS DESCRIPTION AND FLOW DIAGRAM (PFD)

5.1 Introduction

The direct hydration process is recommended for IPA production, operating at 250°C and 150-250 atm [13]. Direct hydration is a one step process where propylene reacts with water to form IPA, with a minor byproduct of DIPE. Figure A1 (enclosed in Appendix 1) shows the process flow diagram (PFD) for the reactor and the first distillation column. Table A1 (enclosed in Appendix 1) shows the stream table for the reactor and the distillation column. The PFD and stream table provide key information for each stream transported from one process unit to another. Two major steps are described: reacting propylene and water to produce IPA and DIPE, and separating IPA from DIPE and unreacted propylene.

5.2 Reacting propylene and water to produce isopropyl alcohol

Propylene and water are heated to 250°C through separate heat exchangers (E-101, E-102). Propylene is compressed to 150 atm using a compressor (C-101). Water is compressed to the same pressure using a pump (P-101). The reactants are mixed in the reactor (R-101) where 70% of the propylene is converted to IPA, and DIPE is formed as a byproduct [13]. IPA forms an azeotrope with water. The fluid mixture exiting the reactor is depressurized by a valve and is cooled using cooling water to its dew point of 81.7°C in a heat exchanger (E-103).

5.3 Separating isopropyl alcohol-water azeotrope and byproducts

The exiting stream enters a distillation column (T-101) operating between 68.4°C and 80.3°C to separate the components of the mixture [17], [18]. This produces a vapour mixture of propylene and DIPE and a liquid consisting of the IPA-water azeotrope. The vapour stream is fed into a condenser (E-104) and reflux drum (V-102). The liquid product from the reflex drum is recycled to the distillation column and the vapour product is enriched in propylene. DIPE is separated in a second distillation column. The liquid IPA stream is split into two: one stream is sent to a reboiler (E-105) and the other stream is processed in an extractive distillation column.

6.0 CONTROL STRATEGY

6.1 Introduction

Control strategies for the input stream of the reactor (R-101) in the production of IPA are described. The control strategies focus on controlling the temperature and pressure of propylene before it enters the reactor. Temperature control ensures propylene enters the reactor at 250°C. Pressure control ensures propylene enters the reactor at 150 atm. A process flow diagram is enclosed.

6.2 Unit controlling temperature

The heat exchanger (E-101) utilizes high-pressure steam to increase the temperature of propylene to 250°C. The temperature in the exiting stream is measured by a temperature transmitter (TT-100). Transmitter TT-100 sends a signal to a temperature controller (TC-100) which opens or closes a control valve (CV-100). Valve CV-100 controls the flow of the steam [21]. Valve CV-100 closes if the temperature in the exiting stream is too high, limiting steam that will transfer heat to propylene. Valve CV-100 opens if the temperature is too low, allowing propylene to be heated up by steam. Refer to Figure E1 (enclosed) in appendix E for details.

6.3 Unit controlling pressure

The compressor (C-101) controls the pressure of propylene in stream 2 followed by a control valve (CV-101) [21]. A pressure transmitter (PT-100) measures the pressure of the stream entering R-101. Transmitter PT-100 sends a signal to a pressure controller (PC-100) which opens or closes CV-101. Valve CV-101 controls the flow of the stream exiting C-101. Valve CV-101 opens if the pressure is too low, allowing gas to flow out of the compressor and increase pressure. Valve CV-101 closes if the pressure is too high, restricting gas flow and reducing pressure. Refer to Figure E1 in appendix E for details.

The high-pressure steam heat exchanger temperature and the compressor pressure for propylene are controlled by the two distinct control loops described.

7.0 FINDINGS AND RECOMMENDATIONS

This report outlines the production of isopropyl alcohol (IPA) at 99.99% purity. IPA is a prevalent chemical used in solvents, cleaners, and cosmetic products. To support growing demand in the Asia-Pacific region, our team aims to produce 100,000 tonnes of IPA at a plant located in Tokuyama, Japan.

The chosen process, direct hydration of propylene, is highly selective and environmentally friendly, producing minimal byproducts. Aqueous phosphotungstic acid is used as a catalyst under high temperature and pressure. The reaction effluent is cooled by a heat exchanger, then separated by distillation.

Two control loops regulate the temperature and pressure of propylene entering the reactor, controlling flow of steam in the heat exchanger and the stream flow through the compressor. Unreacted propylene is recycled to the reactor, and byproducts are burnt as fuel to minimize waste.

Key findings include:

- Direct hydration of propylene produces IPA and DIPE, where IPA forms an azeotrope with water
- IPA-water azeotrope is broken using extractive distillation
- Energy needed to cool down reaction effluent is 7.173 GJ/h

Our recommendations are as follows:

- Use direct hydration of propylene to maximize selectivity and minimize environmental impacts
- Separate products using distillation columns

Additional information is included in the appendices.

REFERENCES

- [1] “Tokuyama Climate, Weather By Month, Average Temperature (Japan) - Weather Spark.” Accessed: Sep. 25, 2024. [Online]. Available: <https://weatherspark.com/y/143098/Average-Weather-in-Tokuyama-Japan-Year-Round>
- [2] “Isopropyl alcohol global market volume 2015-2030 | Statista.” Accessed: Sep. 25, 2024. [Online]. Available: <https://www.statista.com/statistics/1245200/isopropyl-alcohol-market-volume-worldwide/>
- [3] “Isopropyl alcohol - American Chemical Society.” Accessed: Sep. 25, 2024. [Online]. Available: <https://www.acs.org/molecule-of-the-week/archive/i/isopropyl-alcohol.html>
- [4] “Illustrated Glossary of Organic Chemistry - Isopropanol.” Accessed: Sep. 25, 2024. [Online]. Available: <https://www.chem.ucla.edu/~harding/IGOC/I/isopropanol.html>
- [5] J. E. Logsdon and R. A. Loke, “Isopropyl Alcohol,” *Kirk-Othmer Encyclopedia of Chemical Technology*, Dec. 2000, doi: 10.1002/0471238961.0919151612150719.A01.
- [6] “Propylene (Polymer Grade) Prices, Database, News, Chart.” Accessed: Sep. 25, 2024. [Online]. Available: <https://www.procurementresource.com/resource-center/propylene-polymer-grade-price-trends>
- [7] “Isopropyl Alcohol Prices News Monitor Analysis & Demand.” Accessed: Sep. 25, 2024. [Online]. Available: <https://www.chemanalyst.com/Pricing-data/isopropyl-alcohol-31>
- [8] “Isopropyl alcohol - What It Is & How It’s Made | Puracy.” Accessed: Oct. 02, 2024. [Online]. Available: <https://puracy.com/blogs/ingredients/isopropyl-alcohol>
- [9] “Acetone Hydrogenation to Isopropyl Alcohol | Lummus Technology.” Accessed: Oct. 02, 2024. [Online]. Available: <https://www.lummustechnology.com/process-technologies/petrochemicals/styrenics-phenolics-carbonates/styrenics-phenolics-carbonates-production/acetone-hydrogenation-to-isopropyl-alcohol>
- [10] S. Basu and N. C. Pradhan, “Kinetics of acetone hydrogenation for synthesis of isopropyl alcohol over Cu-Al mixed oxide catalysts,” *Catal Today*, vol. 348, pp. 118–126, May 2020, doi: 10.1016/J.CATTOD.2019.07.051.
- [11] A. A. Al-Rabiah, I. Boz, V. M. Akhmedov, M. M. M. Mostafa, and A. A. Bagabas, “Highly Selective Gas-Phase Catalytic Hydrogenation of Acetone to Isopropyl Alcohol,” *Catalysts* 2022, Vol. 12, Page 1251, vol. 12, no. 10, p. 1251, Oct. 2022, doi: 10.3390/CATAL12101251.
- [12] J. Klabunde, C. Bischoff, and A. J. Papa, “Propanols,” *Ullmann’s Encyclopedia of Industrial Chemistry*, pp. 1–14, May 2018, doi: 10.1002/14356007.A22_173.PUB3.

- [13] Y. Onoue, Y. Mizutani, S. Akiyama, Y. Izumi, and H. Ihara, “Isopropyl Alcohol by Direct Hydration of Propylene*”.
- [14] “PROPYLENE | CAMEO Chemicals | NOAA.” Accessed: Oct. 14, 2024. [Online]. Available: <https://cameochemicals.noaa.gov/chemical/4355>
- [15] “Water - Thermophysical Properties.” Accessed: Oct. 14, 2024. [Online]. Available: https://www.engineeringtoolbox.com/water-thermal-properties-d_162.html
- [16] “Isopropyl Alcohol.” Accessed: Oct. 14, 2024. [Online]. Available: <https://webbook.nist.gov/cgi/cbook.cgi?ID=C67630&Mask=4#Thermo-Phase>
- [17] S. Kalla, S. Upadhyaya, K. Singh, R. K. Dohare, and M. Agarwal, “A case study on separation of IPA-water mixture by extractive distillation using aspen plus,” *International Journal of Advanced Technology and Engineering Exploration*, vol. 3, no. 24, pp. 2394–7454, doi: 10.19101/IJATEE.2016.324004.
- [18] “Diisopropyl ether.” Accessed: Oct. 14, 2024. [Online]. Available: <https://webbook.nist.gov/cgi/inchi?ID=C108203&Mask=4>
- [19] “ETHYLENE GLYCOL | CAMEO Chemicals | NOAA.” Accessed: Oct. 14, 2024. [Online]. Available: <https://cameochemicals.noaa.gov/chemical/8660>
- [20] “Chapter 12. Synthesis of the PFD from the Generic BFD | Analysis, Synthesis, and Design of Chemical Processes, Fourth Edition.” Accessed: Oct. 14, 2024. [Online]. Available: <https://learning.oreilly.com/library/view/analysis-synthesis-and/9780132618724/ch12.html#ch12lev1sec1>
- [21] W. Y. . Svrcek, D. P. . Mahoney, and B. R. . Young, “A real time approach to process control,” 2014, Accessed: Dec. 04, 2024. [Online]. Available: <https://www.wiley.com/en-ie/A+Real-Time+Approach+to+Process+Control%2C+3rd+Edition-p-9781119993872>

APPENDICES

Appendix A: Economics calculations

1. Sample calculation of Gross Economic Potential (GEP):

$$GEP = \text{Value of Products} - \text{Value of Feeds, where Value} = \text{Price} \times \text{Amount}$$

Chemical Name	Amount (tonnes/yr)	Price (USD/tonne)	Value (USD)
Propylene	70000	816	57,120,000
Water	30000	1	30,000
Isopropyl Alcohol	100000	1165	116,500,000

$$GEP = \$116,500,000 - (\$57,120,000 + \$30,000) = \$59,350,000$$

2. Sample calculation of input-output diagram:

$$\text{Target IPA output} = \frac{100,000 \text{ tonnes}}{1 \text{ yr}} \times \frac{10^6 \text{ g}}{1 \text{ tonne}} \times \frac{1 \text{ mol}}{60 \text{ g}} = 1.67 \times 10^9 \text{ moles/yr}$$

Based on a 1:1:1 stoichiometric ratio, required propylene input = required water input = target IPA output = $1.67 \times 10^9 \text{ moles/yr}$

$$\text{Required propylene mass} = \frac{1.67 \times 10^9 \text{ moles}}{1 \text{ yr}} \times \frac{42 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ tonne}}{10^6 \text{ g}} = 70,000 \text{ tonnes/yr}$$

$$\text{Required water mass} = \frac{1.67 \times 10^9 \text{ moles}}{1 \text{ yr}} \times \frac{18 \text{ g}}{1 \text{ mol}} \times \frac{1 \text{ tonne}}{10^6 \text{ g}} = 30,000 \text{ tonnes/yr}$$

Appendix B: Material balance sample calculations

1. IPA output

Assuming 8,000 hours of operation per year,

To produce 100,000 tonnes of IPA,

$$\text{Mass flow rate } \dot{m}_{IPA} = \frac{100,000 \text{ tonnes}}{1 \text{ yr}} \times \frac{1 \text{ yr}}{8,000 \text{ h}} = 12.5 \text{ tonnes/h}$$

$$\text{Molar flow rate } \dot{n}_{IPA} = \frac{12.5 \text{ tonnes}}{1 \text{ hr}} \times \frac{1 \text{ tonnemol}}{60 \text{ tonnes}} = 0.208 \text{ tonnemol/h}$$

2. Propylene input/output

Assuming 60% propylene conversion and 1:1 stoichiometric ratio:

$$\dot{n}_{p,in} \times 0.6 = 0.208 \text{ tonnemol/h}$$

$$\dot{n}_{p,in} = 0.347 \text{ tonnemol/h}$$

$$\text{Mass flow rate in } \dot{m}_{p,in} = \frac{0.347 \text{ tonnemol}}{1 \text{ h}} \times \frac{42 \text{ tonne}}{1 \text{ tonnemol}} = 14.6 \text{ tonnes/h}$$

$$\text{Mass flow rate out } \dot{m}_{p,out} = \dot{m}_{p,in} \times 0.4 = 5.85 \text{ tonnes/h}$$

$$\text{Molar flow rate out } \dot{n}_{p,out} = \dot{n}_{p,in} \times 0.4 = 0.139 \text{ tonnes/h}$$

3. Water input/output

Again, assuming 60% water conversion and 1:1 stoichiometric ratio:

$$\dot{n}_{w,in} \times 0.6 = 0.208 \text{ tonnemol/h}$$

$$\dot{n}_{w,in} = 0.347 \text{ tonnemol/h}$$

$$\text{Mass flow rate in } \dot{m}_{w,in} = \frac{0.347 \text{ tonnemol}}{1 \text{ h}} \times \frac{18 \text{ tonne}}{1 \text{ tonnemol}} = 6.25 \text{ tonnes/h}$$

$$\text{Mass flow rate out } \dot{m}_{w,out} = \dot{m}_{w,in} \times 0.4 = 2.49 \text{ tonnes/h}$$

$$\text{Molar flow rate out } \dot{n}_{w,out} = \dot{n}_{w,in} \times 0.4 = 0.139 \text{ tonnes/h}$$

Appendix C: Dew point and energy balance calculations

1. Dew point calculation

$$y_i P = x_i P_i^* \quad (\text{Raoult's Law})$$

$$x_i = \frac{y_i P}{P_i^*}$$

$$\sum_i x_i = 1 \Rightarrow \sum_i \frac{y_i P}{P_i^*} = 1$$

$$P \left(\frac{y_P}{P_P^*} + \frac{y_I}{P_I^*} + \frac{y_W}{P_W^*} \right) = 1 \quad \leftarrow \log P^*(T) = A - \frac{B}{T+C} \quad (\text{Antoine Equation})$$

Assuming constant pressure of 1 bar, Given constants A, B, C: (source - NIST)

$$\frac{0.287}{P_P^*(T)} + \frac{0.285}{P_I^*(T)} + \frac{0.429}{P_W^*(T)} = 1$$

$$T = 354.85 \text{ K} \\ = 81.7^\circ \text{C}$$

	A	B	C
Propylene	3.97488	795.819	-24.884
IPA	5.0768	1659.793	-45.854
Water	4.8610	1357.427	-75.814

2. Energy requirements

Energy needed to heat up a gas $\Delta H = \int_{T_{\text{initial}}}^{T_{\text{final}}} C(T) dT$, where $C(T)$ is the molar heat capacity of the gas as a function of temperature.

$C(T) = a + bT + cT^2 + dT^3$, where a, b, c, and d are constants specific to the gas.

The final equation to calculate energy required for 1 mole of each gas component is:

$$\text{Energy (J/mol)} \Delta H = \int_{T_{\text{initial}}}^{T_{\text{final}}} (a + bT + cT^2 + dT^3) dT$$

This equation is used to calculate energy requirements in the spreadsheet below:

IPA						
A	B	C	D	E	F	G
48.7560898	-0.096083222	0.00142278	-2.86E-06	2.69E-09	-1.24E-12	2.26E-16
deltaT	T1	T2 (dew point)				
-176	523	354.85				
integral						
A*deltaT	-8581.071805					
B/2 * (T2^2-T1^2)	7091.44514					
C/3 * (T2^3-T1^3)	-46654.60629					
D/4 * (T2^4-T1^4)	4.21E+04					
E/5 * (T2^5-T1^5)	-1.80E+04					
F/6 * (T2^6-T1^6)	3.82E+03					
G/7 * (T2^7-T1^7)	-3.22E+02					
deltaH	-20535.32916	J/mol				
Propylene						
A	B	C	D	E	F	G
43.22371801	-0.115723958	0.001085358	-2.02E-06	1.83E-09	-8.25E-13	1.48E-16
deltaT	T1	T2 (dew point)				
-176	523	354.85				
integral						
A*deltaT	-7607.37437					
B/2 * (T2^2-T1^2)	8541.034349					
C/3 * (T2^3-T1^3)	-35590.14758					
D/4 * (T2^4-T1^4)	2.97E+04					
E/5 * (T2^5-T1^5)	-1.22E+04					
F/6 * (T2^6-T1^6)	2.54E+03					
G/7 * (T2^7-T1^7)	-2.12E+02					
deltaH	-14824.63172	J/mol				
water						
A	B	C	D	E	F	G
33.17438187	-0.003246334	1.74E-05	-5.98E-09	0	0	
deltaT	T1	T2 (dew point)				
-176	523	354.85				
integral						
A*deltaT	-5838.691209					
B/2 * (T2^2-T1^2)	239.5964559					
C/3 * (T2^3-T1^3)	-5.72E+02					
D/4 * (T2^4-T1^4)	8.81E+01					
E/5 * (T2^5-T1^5)	0					
F/6 * (T2^6-T1^6)	0					
G/7 * (T2^7-T1^7)	0					
deltaH	-6082.714855	J/mol				

$$\begin{aligned}
 \text{Rate of energy loss in reaction effluent } \dot{Q} \text{ (J/h)} &= \sum_i \Delta H_i \text{ (J/mol)} \times \dot{n}_i \text{ (mol/h)} \\
 &= -14825 \times 1.39 \times 10^5 - 6083 \times 1.39 \times 10^5 - 20535 \times 2.08 \times 10^5 \\
 &= -7172723723 \text{ J/h} \\
 &= -7.173 \text{ GJ/h}
 \end{aligned}$$

Total utility required = 7.173 GJ/h

3. Calculations for total utilities and cost

We can start by finding the flow rate of water, $\dot{n}_{water}$, required to remove 7.173 GJ/h of energy.

$$\text{By } \dot{Q} = \dot{n}c\Delta T, \dot{n}_{water} = \frac{7.173 \times 10^9 \text{ J/h}}{76.39 \text{ J/mol K} \times 5 \text{ K}} = 1.88 \times 10^7 \text{ mol} = 18.8 \text{ tonnemol}$$

$$\dot{m}_{water} = \dot{n}_{water} \times M_{water} = \frac{18.8 \text{ tonnemol}}{1 \text{ h}} \times \frac{18 \text{ tonne}}{1 \text{ tonnemol}} = 338 \text{ tonnes/h}$$

$$\text{Cost (\$/h)} = \text{Total energy (GJ/h)} \times \text{cost of cooling (\$/GJ)}$$

$$= \frac{7.173 \text{ GJ}}{1 \text{ h}} \times \frac{\$0.378}{1 \text{ GJ}} = \$2.71/\text{h}$$

$$\text{Cost (\$/yr)} = \frac{\$2.71}{1 \text{ h}} \times \frac{8000 \text{ h}}{1 \text{ yr}} = \$21690/\text{yr}$$

Appendix D: Process Flow Diagram

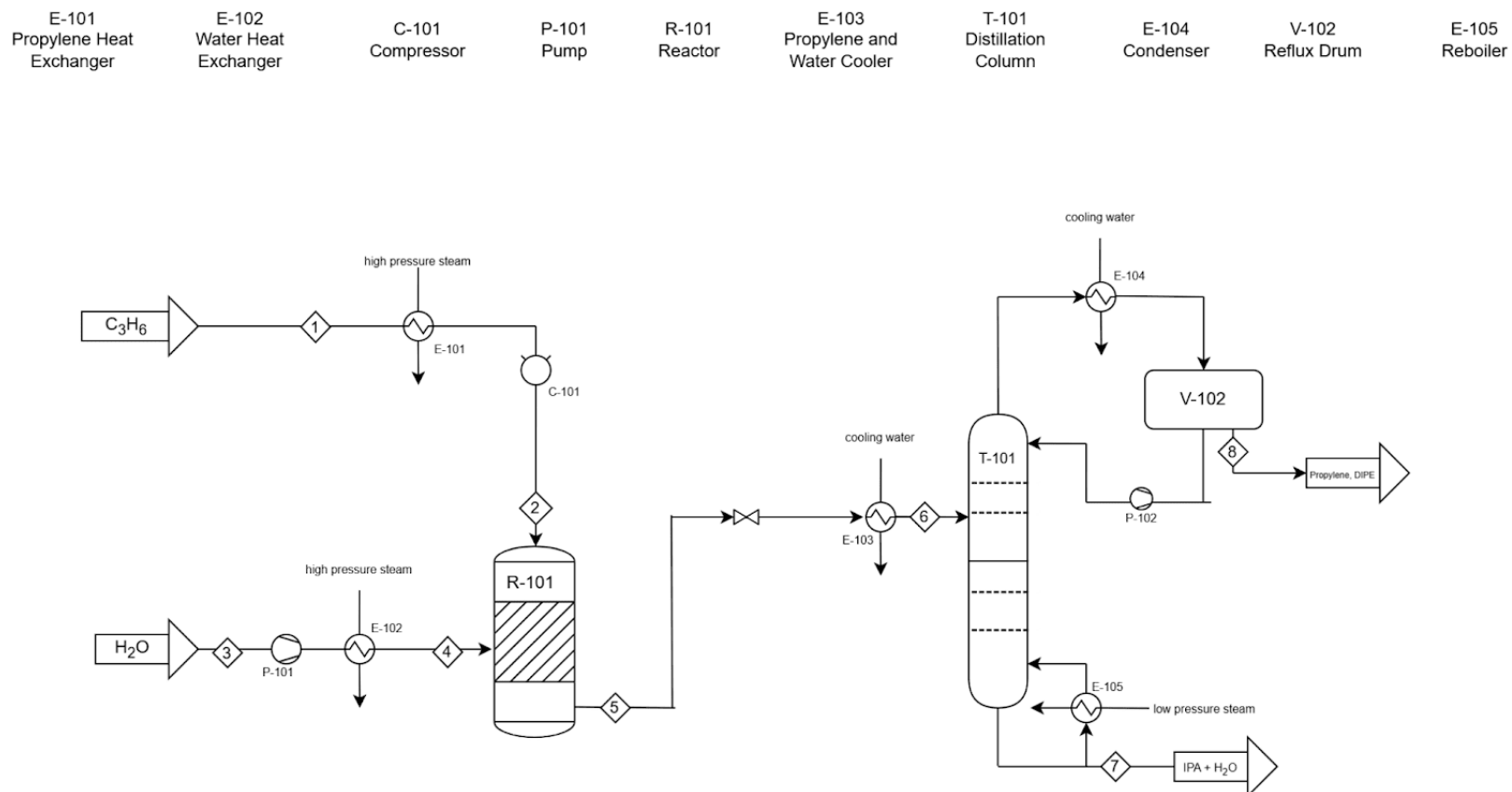


Figure D1: Process Flow Diagram of reactor and distillation column in isopropyl alcohol production

Table D1: Stream table for the reactor and distillation column

Stream Number	1	2	3	4	5	6	7	8
Temperature (°C)	25	250	25	250	250	81.7	68.4	80.3
Pressure (atm)	1	150	1	150	150	1	1	1
Vapour Fraction	1	1	0	0	1	1	0	1
Mass Flow (tonne/h)	14.6	14.6	6.25	6.25	20.85	20.85	14.99	5.86
Molar Flow (tonne-mol/h)	0.347	0.347	0.347	0.347	0.486	0.486	0.346	0.140
Component Molar Flow (tonne-mol/h)	-	-	-	-	-	-	-	-
Propylene	0.347	0.347	0	0	0.139	0.139	0	0.139
Water	0	0	0.347	0.347	0.138	0.138	0.138	0
IPA	0	0	0	0	0.208	0.208	0.208	0
DIPE	0	0	0	0	0.001	0.001	0	0.001

Assumptions used for molar flow calculations:

- Assume 60% conversion rate of propylene and water

Appendix E: Control Strategy Process Flow Diagram

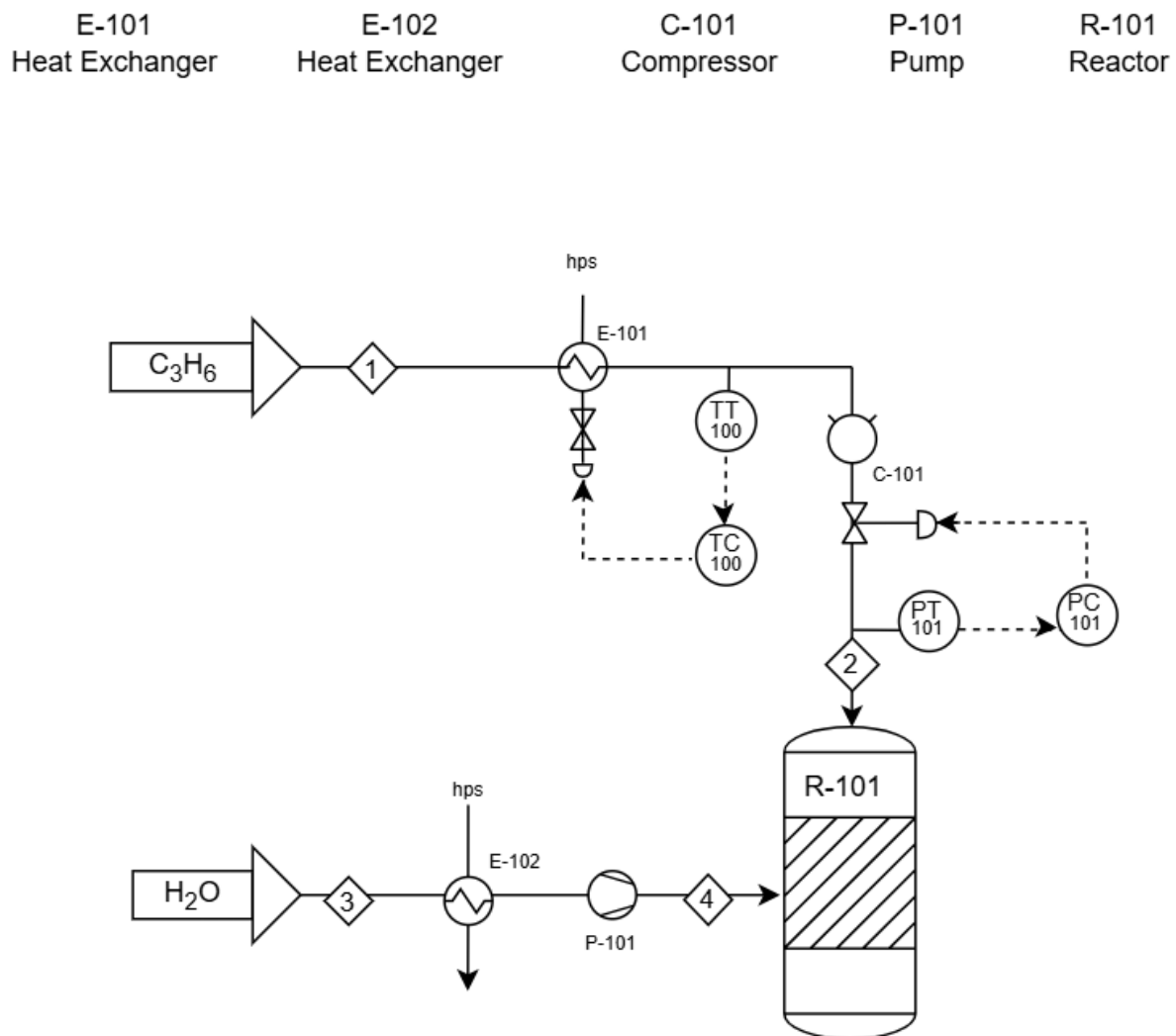


Figure E1: Process flow diagram showing controls for propylene heat exchanger and compressor.